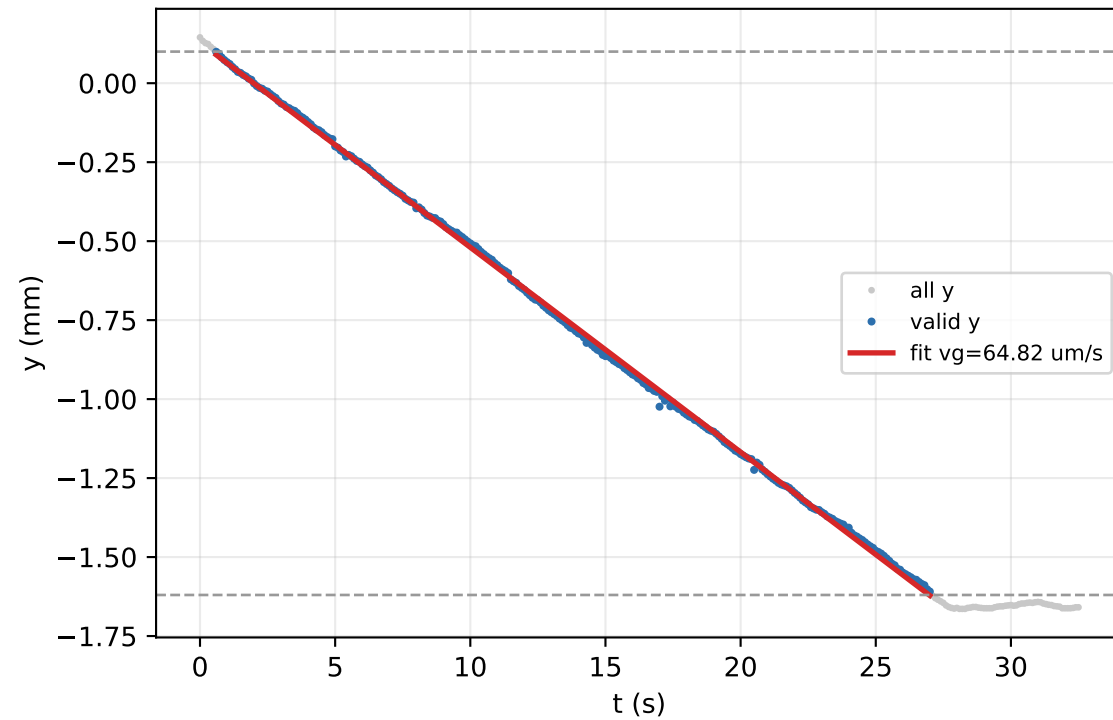
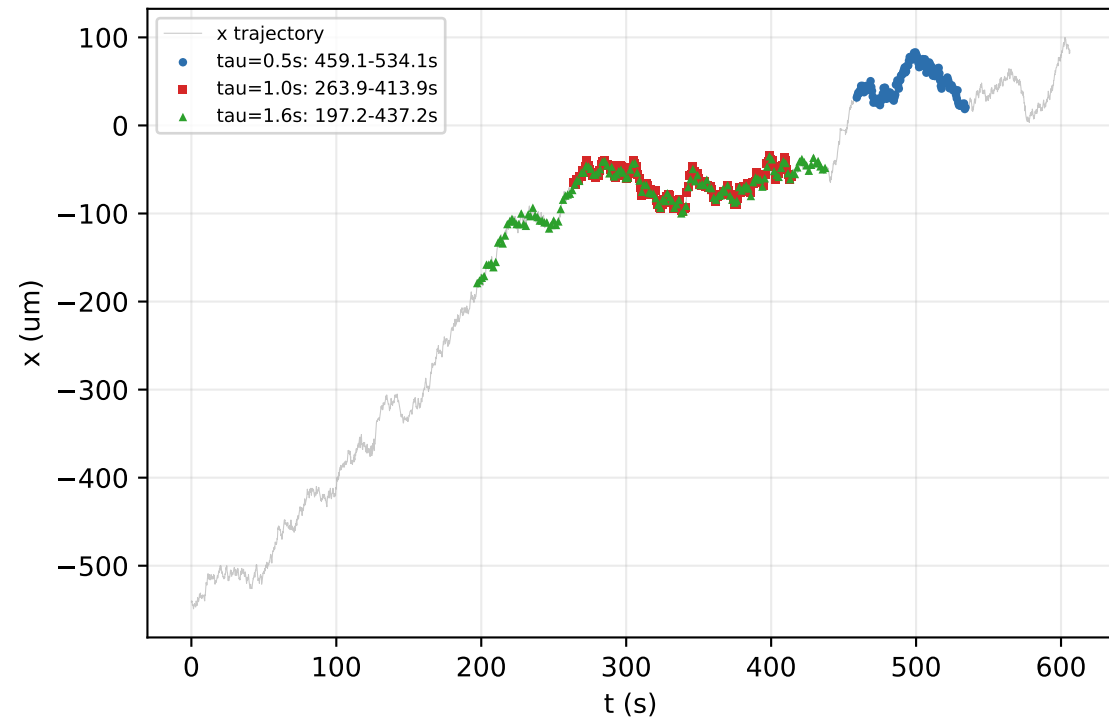


Millikan oil drop particle4: optimized Brownian intervals

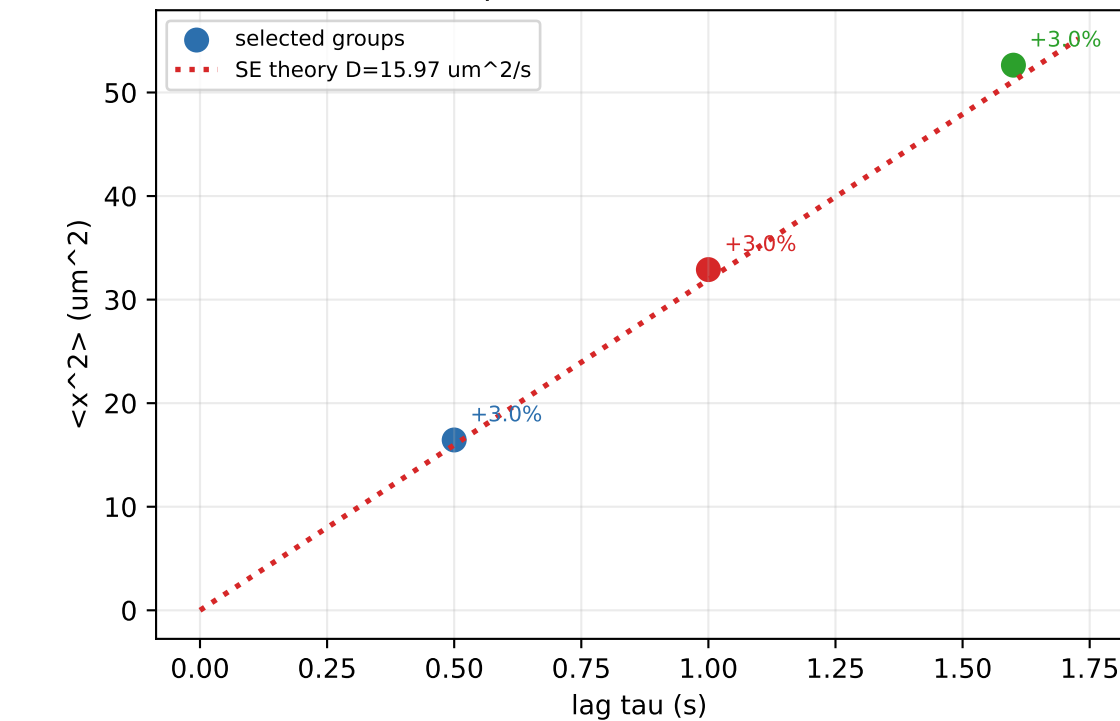
Free-fall terminal velocity



Sampled coordinate points per lag group



Separate $\langle x^2 \rangle$ results



Summary

Particle: particle4
 Window strategy: target_kb
 Freefall valid points: 265
 $vg = 64.8226 \text{ um/s}$, $R^2 = 0.999592$
 $r_{\text{eff}} = 0.7452 \text{ um}$, $d_{\text{eff}} = 1.4905 \text{ um}$
 SE theory $D = 15.9676 \text{ um}^2/\text{s}$

τ	covered time	$\langle x \rangle$	$\langle x^2 \rangle$	k_B err
0.5s	459.1-534.1s	-0.076	16.447	+3.00%
1.0s	263.9-413.9s	0.044	32.898	+3.01%
1.6s	197.2-437.2s	0.872	52.638	+3.02%

NIST $k_B = 1.3806\text{e-}23 \text{ J/K}$
 No average is reported.